

7.6.4 Annunciators

Fuel system operation is monitored by the following annunciators:

- **LOW FUEL warning light.** A fuel level float switch located in the supply tank activates the **LOW FUEL** warning light when the supply tank usable quantity is below 60 kg.
- **FILT 1, FILT 2 warning lights.** The **FILT 1 (2)** warning light will be activated when the engine-mounted filter becomes clogged.

7.7 POWER TRAIN (see Fig. 7-9)

The power train transmits engine power to both rotors and the transmission-mounted accessories. The power train comprises two engine drive shafts, a main transmission with accessory drives, tail rotor drive shafts, an intermediate gearbox, and a tail rotor gearbox.

7.7.1 Main Transmission (see Fig. 7-9)

Mounted forward of the engines above the cargo compartment, the main transmission combines the two engine drive shaft inputs and provides drive to the main rotor, tail rotor, accessories, and oil cooler fan. Two freewheeling units at the main transmission inputs permit either engine to be disengaged from the transmission during single-engine operation or both engines during autorotation.

Power from the engines is transmitted through various stages of reduction gearing to obtain the necessary speed for the main rotor and tail rotor. Two hydraulic pumps and a rotor rpm (N_{R0}) tachgenerator are driven by the accessory gearbox mounted on the transmission left side. Mounted on the transmission right side is an oil cooler blower fan and reduction gearbox assembly.

7.7.2 Main Transmission Oil System

The main transmission oil system lubricates and cools transmission components. An oil pump mounted on the bottom right of the transmission, suctions oil from the sump through a filter screen and feeds it then to a fine-mesh filter mounted on the forward side of the oil cooler assembly. The filter incorporates a bypass to ensure oil circulation in the event of filter clogging. Oil from the filter then flows to the oil cooler which is fitted between the two engine oil coolers. The cooled oil is then returned to a transmission case manifold fitting located on the aft side below the tail rotor drive and between the two input drives. An external line from the manifold fitting supplies oil for the pressure sensors. Passages within the transmission casings distribute oil to jets used in spraying critical components.

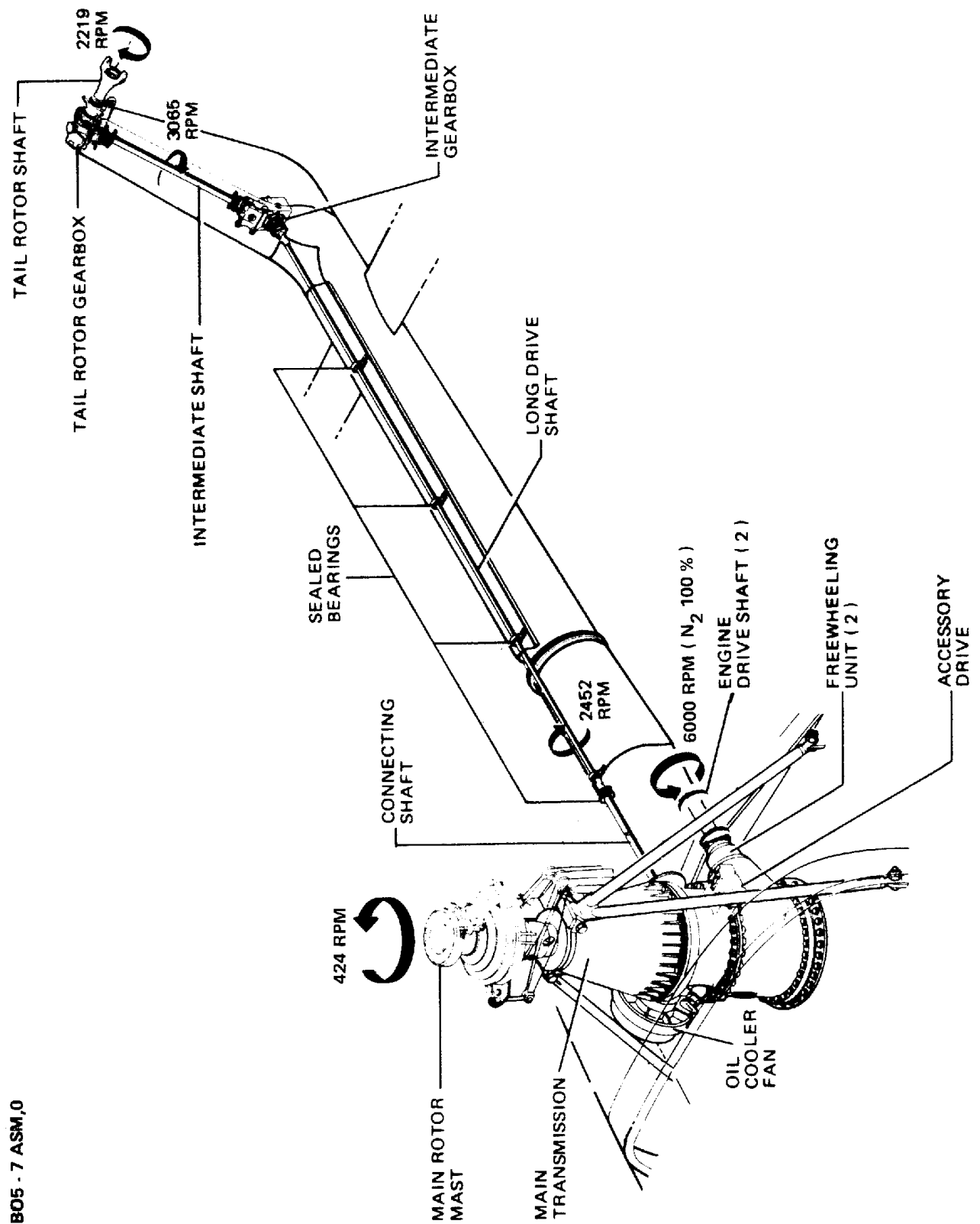


Fig. 7-9 Power Train

7.7.3 Main Transmission Indicators (see Fig. 7-10)

Triple tachometer indicator (ENG-ROTOR PERCENT RPM) indicates rotor speed (N_{Ro}) on an inner scale in percent rpm. The rotor rpm signals are produced by a tachgenerator mounted on the aft face of the transmission accessory drive.

Triple oil temperature indicator (°C TEMP OIL) upper scale labeled R, indicates the transmission oil temperature. The temperature sensor is mounted on a transmission oil pump outlet manifold.

Triple oil pressure indicator (kp/cm² PRESS OIL) upper scale labeled R, indicates the transmission oil pressure. The oil pressure transmitter is installed in a T-fitting secured to an engine oil line hose clamp mounted on the transmission compartment floor.

7.7.4 Main Transmission Annunciators (see Fig. 7-10)

T. OIL warning light will be activated by either an overtemperature switch installed in the transmission case below the cooler fan and/or by a pressure switch installed in the T-fitting adjacent to the oil pressure transmitter. The overtemperature switch closes at an oil temperature over 105 °C while the pressure switch closes whenever the oil pressure is below 0.5 kp/cm².

EFFECTIVITY 0451-9999, and 0001-0450 After SB 60-37

OIL COOL warning light monitors operation of the oil cooler blower fan via a differential air pressure switch mounted on the oil cooler air inlet duct above the blower fan. A hose connects the switch to the air outlet shroud between the fan and oil cooler assembly. The pressure switch will activate the warning light whenever the pressure increase caused by the fan drops below a certain value.

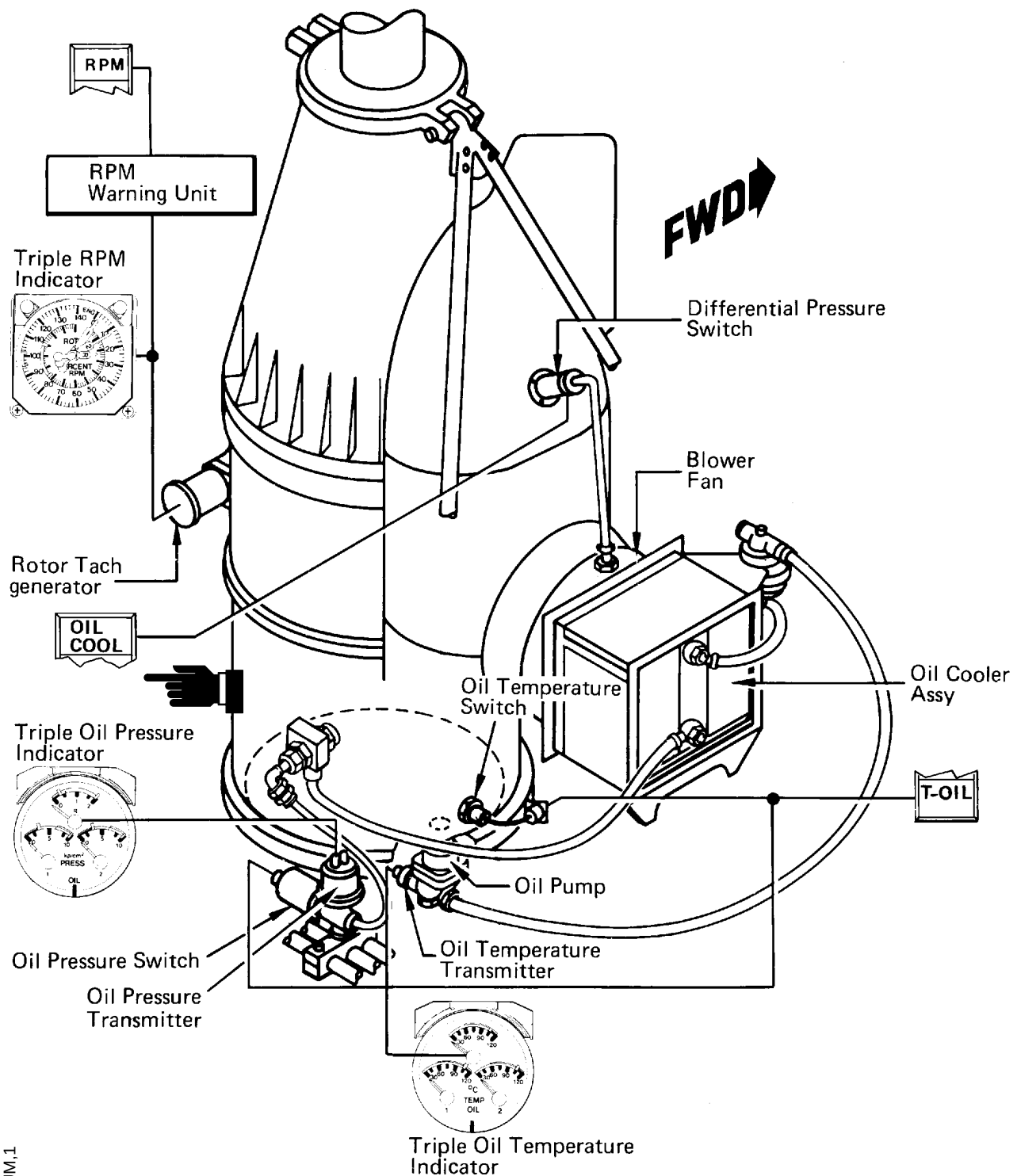
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7.7.5 Mast Moment Indication System

The mast moment indication system measures and indicates bending moments of the main rotor mast. The system is comprised of a strain gauge bridge, a rotary induction transmitter and a mast moment indicator.

Four strain gauges are bonded to the upper inner surface of the hollow rotor mast and are combined to form a bridge circuit. The rotary induction transmitter that conveys signals to and from the bridge circuit is located at the base of the main transmission.

The mast moment (MM) indicator is located on the instrument panel and is comprised of an indicator needle with colored scale, a calibration (CAL) screw, a TEST pushbutton switch and a **LIMIT** warning light.



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Fig. 7-10 Main Transmission Monitoring Systems

The MM indicator is powered from the Main Bus via a fuse. The indicator generates a carrier frequency which is supplied to the strain gauge bridge via the rotary transmitter. Bending moments of the rotor mast alter the resistance value of the strain gauge bridge. This modified signal is then returned through the rotary transmitter to the MM indicator where it is evaluated and then indicated by a needle deflection.

The system can be checked with the TEST pushbutton provided. When electrical power is initially applied to the system, the **LIMIT** warning light may come on. Pressing the TEST pushbutton should cause **LIMIT** warning light to illuminate and the needle to deflect to the white calibration line. Releasing the button will cancel the test indications, i.e. the **LIMIT** warning light goes off and the needle returns to the zero point (white line).

The Mast moment warning light begins to flash when the needle reaches the end of the yellow arc. It illuminates steadily when reaching the red radial. It keeps steady on at 95% of full deflection (just before reaching the white radial) even if mast moment will be reduced. It can be reset by pressing the TEST push button.

7.7.6 Tail Rotor Drive Shafts (see Fig. 7-9)

There are three tubular tail rotor drive shaft sections comprising:

- a connecting shaft with flexible couplings,
- a long shaft supported by sealed bearings and fitted with a flexible coupling at the intermediate gearbox end,
- an intermediate shaft with flexible couplings installed at the vertical fin between the intermediate and tail rotor gearboxes.

The connecting shaft is routed through a tunnel in the engine compartment dividing fire-wall. The long drive shaft and the intermediate shaft are covered by fairings which may be opened for maintenance and inspection.

7.7.7 Intermediate Gearbox (see Fig. 7-9)

The intermediate gearbox mounted at the base of the vertical fin increases the main transmission output rpm and changes the angle of drive 50°. The gearbox housing provides for mounting of a tail rotor yaw control bell crank. This gearbox is wet sump lubricated with an oil level sight gauge located on the left side. A magnetic-type drain plug is installed in the base of the gearbox housing.

7.7.8 Tail Rotor Gearbox (see Fig. 7-9)

The tail rotor gearbox mounted on top of the vertical fin reduces the output rpm and changes the angle of drive 90°. The tail rotor shaft passes through and is supported by the gearbox output bevel gear. The gearbox housing provides for mounting of a tail rotor yaw control bell crank. The tail rotor gearbox is also wet sump lubricated with an oil level sight gauge located on the rear face of the gearbox housing. A magnetic-type drain plug is installed at the lower rear of the housing.